

Expectation Decider

1) Probability

⇒ Probability is the measure of how likely an event is to occur.

$$P(E) = \frac{\text{Favourable outcomes}}{\text{Total outcomes}}$$

0 = Impossible

1 = Certain

0.5 = 50% chance

Example :-

Total Students 200

Passed Students 120

$$P(\text{Pass}) = \frac{120}{200} \Rightarrow 0.6$$

$$P(\text{Pass}) = 0.6 = 60\%$$

2) Probability Terminology

1) Experiment

An action or process that produce an outcome.

Example : Selecting one student randomly.

2) Outcome

Outcome is result of an experiment.

Example : Selected student either passes or fail.

3) Sample Space

The set of all possible outcomes.

Example : $S = \{\text{pass, fail}\}$

4) Event

5) Favourable outcome

3) Probability Event example

Event A - Student Passes

$$P(A) = \frac{120}{200} = 0.6 = 60\%$$

Event B - Students fails

$$P(B) = \frac{80}{200} = 0.40 = 40\%$$

Event C - Students has attendance > 80%
90 Students with attendance above 80%

$$P(C) = \frac{90}{200} = 0.45 = 45\%$$

2) Types of Events

1) Empirical Probability

Empirical Probability is calculated from actual observed data.

Example :-

$$P(E) = \frac{\text{Number of times event occurs}}{\text{Total observations}}$$

Passed Students = 120

Total Students = 200

$$P(E) = \frac{120}{200} = 0.60 \text{ or } 60\%$$

Empirical Probability of Passing = 60%

2) theoretical Probability

theoretical Probability is based on mathematical assumptions rather than observed experimental results.

Suppose,

I have Selected 3 Students from the Total Students 200.

Let, $X = \text{Students who Passes}$

Total = 200

Pass = 120

Fail = 80

3 Selected

$$P(X=2) = \frac{\binom{120}{2} \binom{80}{1}}{\binom{200}{3}}$$

$$P(X=2) = 0.4349$$

2) 43%

3 Random Variable

Random Variable is a Variable whose value depends on the outcome of a random experiment.

Example

I randomly Selected 3 Students

Let,

$X =$ Number of Students who Pass.

So the possible values

$$X = \{0, 1, 2, 3\}$$

Selected Students

(Pass, Pass, fail)

So, $X = 2$

then the random Variable $X = 2$

2) Probability distribution table

A Probability distribution Shows all Possible values of a random variable and their corresponding probability.

X

meaning

0

No Student Passes

1

1 Student Passes

2

2 Students Passes

3

ALL 3 Student Passes

this Probability distribution are calculated using the hypergeometric Distribution.

formula :
$$P(X=x) = \frac{\binom{K}{x} \binom{N-K}{n-x}}{\binom{N}{n}}$$

Teacher's Signature _____

3) mean of random variable

the mean of a random variable is its expected value or long run average outcome.

Formula : $E(x) = \sum x P(x=x)$

If 3 Students are selected randomly we expect 1.8 Students on average to Pass.

★ $\Rightarrow$ Variance of random variable

Variance measures how much the values of a random variable vary around its mean.

Formula : $Var(x) = E(x^2) - [E(x)]^2$

So the variance of the number of Passing Students among 3 Selected Students is approximately 0.7128.

4 Venn Diagram

Event A

Students studying more than 10 hours/week.

Event B

Students having attendance $> 80\%$

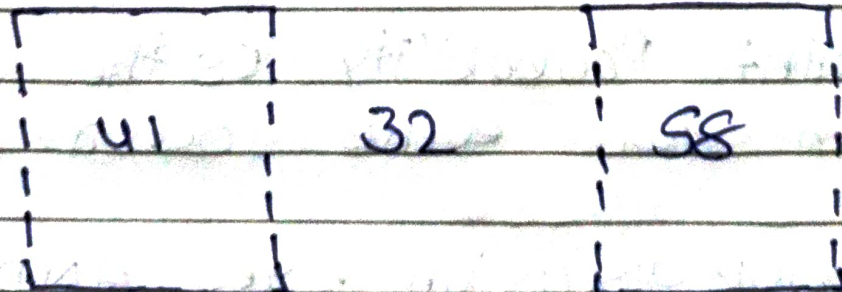
$A = 73$ Students

$B = 90$ Students

$A \cap B = 32$ Students

A

B



32 Students both:

- Study more than 10 hours/week.
- Students have attendance above 80%.

Teacher's Signature _____

5 contingency table

contingency table two categorical variables ke relationship ko show karti hai.

group discussion x final exam result

group discussion	Pass	fail	All
yes	85	36	121
no	44	35	79
All	129	80	209

★ Joint Probability

Joint Probability is the probability that two events occur together

group discussion = yes AND Pass

Students = 85

Total = 200

$$P(\text{UNPASS}) = \frac{85}{200} = 0.425$$

$$\Rightarrow 42.50\%$$

★ marginal probability

marginal probability is the probability of a single event without considering another variable.

Probability of Passing.

$$P(\text{Pass}) = \frac{120}{200} = 0.60 = 60\%$$

★ Conditional Probability

Conditional Probability is the probability of one event occurring given that another event has already occurred.

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Probability of Passing given that Student Participated in group discussion.

$$P(\text{Pass}/\text{GD}) = 70.25\%$$

Students who Participated in group discussions the Probability of Passing was 70.25%

conclusion ∴ This is the higher than the overall Passing Probability of 60% .

6 Independent Events

Two events are independent if the occurrence of one event does not affect the probability of the other.

★ Dependent Events.

Two events are dependent if the occurrence of one event changes the probability of the other.

Example :

$$P(\text{Pass}) = 60\%$$

$$\text{But, } P(\text{Pass/CD}) = 70.25\%$$

$$\text{So, } 60\% \neq 70.25\%$$

∴ Group Discussion and Passing are Dependent Events.

★ mutually Exclusive events

Two events are mutually exclusive if they cannot occur ~~at the~~ at the same time.

Example : Group discussion and Passing are NOT mutually exclusive.

Because, a student can participate in group discussion + Pass

≠ Bayes' theorem

Baye's theorem is used to calculate the probability of an event based on prior information or evidence

$$P(A/B) = \frac{P(B/A)P(A)}{P(B)}$$

High attendance $\geq$ Attendance $> 80\%$

$$P(\text{Pass} / \text{High attendance}) = 77.78\%$$

also calculated from collab file.

$$P(\text{Pass}) = 66.67\%$$

According to the given historical information, a Student with high attendance has a 77.78% Probability of Passing:

8) Final Conclusion

The Expectation Decided Project analyzed the academic performance of 200 students using different probability and statistical technique.

The dataset showed that 120 students pass and 80 students failed, giving an empirical passing probability of 60%.

A random variable X was defined as the number of passing students among 3 randomly selected students. Using the Hypergeometric Distribution the expected value was found to be 1.8 and the variance was approximately 0.7128.

The Venn diagram showed 73 students studied more than 10 hours/week, 90 students had attendance above 80%, and 32 students satisfied both conditions.

The conditional probability analysis showed that the probability of passing given group discussion was 70.25%, compared with the overall passing probability of 60%. Therefore, group discussion participation and passing were considered dependent events in the dataset.

Bayes' theorem showed a 77.78% probability of passing given high attendance based on the historical probabilities used in the project.

The End.